

Note 5.5 Orthonormal Sets

Intro

Definition of Orthogonal Set

For a set $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$, the inner product is zero of each pair of different vectors. And the set is orthogonal set.

Theorem 5.5.1

The vectors in an orthogonal set without zero vectors are linearly independent.

Definition of Orthonormal Set

An orthogonal set of unit vectors.

It can be obtained from orthogonal set by dividing the norms of the vectors.

And If the set is a basis of a subspace, it is called orthonormal basis.

Theorem 5.5.2

For an orthonormal basis for V $\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$, if $\mathbf{v} = \sum_{i=1}^n c_i \mathbf{u}_i$, then $c_i = \langle \mathbf{v}, \mathbf{u}_i \rangle$.

c_i is the coordinates of $\mathbf{v}$ with respect to the basis set $\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$.

It can be proved by multiply $\mathbf{u}_i$ of both sides.

Corollary 5.5.3

For an orthonormal basis for V $\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$, if $\mathbf{u} = \sum_{i=1}^n a_i \mathbf{u}_i$ and $\mathbf{v} = \sum_{i=1}^n b_i \mathbf{u}_i$, then

$$\langle \mathbf{u}, \mathbf{v} \rangle = \sum_{i=1}^n a_i b_i$$

Corollary 5.5.4 Parseval's Formula

For an orthonormal basis for V $\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$ and $\mathbf{v} = \sum_{i=1}^n c_i \mathbf{u}_i$, then

$$\|\mathbf{v}\|^2 = \sum_{i=1}^n c_i^2$$

Orthogonal Matrices

Definition of Orthogonal Matrix

An $n \times n$ matrix Q is a orthogonal matrix if the column vectors of Q form an orthonormal basis in $\mathbb{R}^n$.

That is, the column vectors

- linearly independent
- unit vector

Theorem 5.5.5

An $n \times n$ matrix Q is orthogonal if and only if $Q^T Q = I$. And $Q^{-1} = Q^T$.

Multiplication by an orthogonal matrix preserves the lengths of vectors.

Properties of Orthogonal Matrices

- The column vectors of Q form an orthonormal basis for $\mathbb{R}^n$.
- $Q^T Q = I$
- $Q^T = Q^{-1}$
- $\langle Q\mathbf{x}, Q\mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle$
- $\|Q\mathbf{x}\| = \|\mathbf{x}\|$

Permutation Matrices

Definition of Permutation Matrices

A permutation matrix is formed from the identity matrix by reordering its columns. And it is orthogonal matrices.

Orthonormal Sets and Least Squares

Theorem 5.5.6

If the column vectors of A form an orthonormal set of vectors in $\mathbb{R}^m$ (not orthogonal matrix since not necessary square matrix), the solution of the least squares problem is

$$\hat{\mathbf{x}} = A^T \mathbf{b}$$

since $A^T A = I$.

Theorem 5.5.7

For an orthonormal basis for S which is the subspace of V $\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$, if

$$\mathbf{p} = \sum_{i=1}^n c_i \mathbf{u}_i$$

where $c_i = \langle \mathbf{x}, \mathbf{u}_i \rangle, \mathbf{x} \in V$,
then

$$\mathbf{p} - \mathbf{x} \in S^\perp$$

In fact, $\mathbf{p}$ is the projection of $\mathbf{x}$ onto S .

Theorem 5.5.8

Under theorem 5.5.7, $\mathbf{p}$ is the element of S that is closest to $\mathbf{x}$.

Corollary 5.5.9

Let S be a **nonzero subspace** of $\mathbb{R}^m$ and let $\mathbf{x} \in \mathbb{R}^m$, $U = (\mathbf{u}_1, \dots, \mathbf{u}_n)$ where the set of column vectors is the orthogonal basis for S , then the projection $\mathbf{p}$ of $\mathbf{x}$ onto S is

$$\mathbf{p} = UU^T \mathbf{x}$$